Yunrui Guo

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Summary

AI researcher and system engineer focused on building physically grounded AI systems that move from perception to reasoning to action. My strongest fit is bridging foundation-model reasoning, structured task planning, robotics simulation, and real-world deployment. Driven by strong technical curiosity, I combine research depth with full-stack engineering ownership across embodied decision-making, multi-agent collaboration, generative modeling, visual perception and understanding. The aim is to turn foundation models into executable, verifiable, and deployable intelligence for autonomous platform, aligned with real-world product needs, platform ecosystems, and future AI applications. AI researcher and system engineer focused on building physically grounded AI systems that move from perception to reasoning to action. My strongest fit is bridging foundation-model reasoning, structured task planning, robotics simulation, and real-world deployment. Driven by strong technical curiosity, I combine research depth with full-stack engineering ownership across embodied decision-making, multi-agent collaboration, generative modeling, visual perception and understanding. The aim is to turn foundation models into executable, verifiable, and deployable intelligence for autonomous platform, aligned with real-world product needs, platform ecosystems, and future AI applications.

Work Experience

Senior AI Engineer | Intelligent Game and Decision Lab, Beijing Jan 2022 – Dec 2025

- Led the design and implementation of a multimodal intelligent decision platform integrating textual, visual, and structured state information for autonomous reasoning and decision-making. Developed AI-agent workflows for strategic reasoning, cooperative/competitive behaviors, and adaptive policy evolution in dynamic environments.
- Managed a 12-engineer development team across architecture, model training, simulation integration, optimization, testing, documentation, and delivery.
- Supervise doctoral students and support academic publications and project outcomes.

Visiting Ph.D. | University of Padova, VIMP Group | Italy Oct 2019 – Apr 2021

- Conducted research on computer vision, representation learning, and open-set recognition under Prof. Alessandro Sperduti and Prof. Lamberto Ballan http://vimp.math.unipd.it
- Developed the Conditional Variational Capsule Network for open-set recognition, published at IEEE/CVF ICCV 2021.

Projects

Multimodal Intelligent Decision Platform Jan 2022 – Oct 2025

Architected and implemented a simulator-agnostic embodied decision platform that converts structured world states, textual instructions, and visual observations into executable high-level trajectories. The Decision Pipeline stacks multimodal inputs → LLM-generated course of action → schema/action normalization → precondition and reachability validation → optional rollout/MCTS-lite → executable trajectory to mitigates local vLLM/Qwen limitations in long-horizon task reasoning and spatial-relation understanding.

Integrated Isaac Lab, AI2-THOR, local vLLM/Qwen serving, and official EmbodiedBench EB-ALFRED custom wrappers without modifying benchmark core code. In a 25-task internal EB-ALFRED-style benchmark, improved executable task success from 8/25 to 25/25; on a representative official EB-ALFRED task, reduced invalid LLM actions from 10 to 0 and achieved 100% task success.
• Filed 2 patent and registered software copyright.

LLM-Agent-Orchestrated Molecular Discovery Workflow May 2025 – June 2026
Developed a research automation workflow for property-guided molecular generation, where an LLM agent coordinates latent diffusion sampling, teacher-classifier guidance, molecular validity checking, QED evaluation, fingerprint-based diversity analysis, t-SNE / DBSCAN clustering, iterative refinement, and structured experiment reporting. The system reformulates molecule generation from one-shot sampling into a closed-loop discovery process, supporting candidate screening, failure-mode diagnosis, condition adjustment, and interpretable structure-property analysis. The workflow enables traceable candidate selection, diversity-aware evaluation, and reproducible refinement across property-oriented molecular generation tasks.
• Manuscript Submitted to Nature Machine Intelligence.

Interpretable Generative Modeling for High-Dimensional Sparse 3D Volumetric Data Oct 2024 - Apr 2026
Developed an interpretable generative framework for high-dimensional sparse 3D volumetric data, addressing training bottlenecks caused by large voxel grids, extreme spatial sparsity, and foreground-background imbalance. Designed a normalization-free VAE with spatially weighted reconstruction loss to learn compact sparse representations, preventing sparse-signal collapse while reducing training time, inference latency, and memory consumption. Trained a conditional latent diffusion model in the compact latent space to enable controllable structure generation under anatomical or structural constraints without directly diffusing over high-dimensional voxel volumes. Developed a closed-form geometric decoder that extracts structured parameters from generated voxel-cluster statistics, improving symbolic interpretability, robustness to sampling noise, and downstream usability. Applied Grad-CAM and UMAP/t-SNE to analyze latent-space organization, feature attribution, and representation separability across structural variations.
• Manuscript under review.

Edge-Device Collaborative Recognition Framework July 2024 – Sep 2024
Proposed a bandwidth-efficient edge-device collaborative recognition framework that combines lightweight on-device models with edge-level foundation models for hierarchical verification. Designed ROI-level compressed feature-vector transmission and confidence-aware edge escalation, allowing high-confidence samples to be processed locally while ambiguous cases are selectively verified by edge foundation models. Explored teacher-aligned compact embeddings and codebook-based vector compression to reduce communication overhead while preserving recognition accuracy.
• Filed 1 patent and registered software copyright.

Dual-Arm Robotic Sorting System for Corrugated Boxes Jan 2024 – Jun 2024
Developed a dual-arm collaborative robotic system for intelligent sorting and manipulation of corrugated boxes in warehouse logistics. Built a **primitive action library** covering grasping, placing, opening, rotation, and dual-arm coordinated manipulation, and implemented **foundation-model-driven task decomposition and orchestration** to compose executable workflows from high-level instructions, achieving **90%** task-decomposition accuracy. Implemented coordinated motion planning and control algorithms for synchronized handling of large boxes, and optimized the grasping model to achieve an overall grasp success rate above **90%**. Built a complete simulation-to-reality pipeline using **Isaac Sim** and **ROS2**, integrating **3D visual localization** and adaptive task planning for stable, high-precision dual-arm operation in industrial warehouse environments.
• Successfully deployed in real warehouse operations.

Hybrid Multi-Sensor Fusion Framework for UAV Swarms June 2023 – Dec 2023
Developed a hybrid multi-sensor fusion framework for UAV swarm-based object search and localization, integrating homogeneous and heterogeneous visual streams across multiple aerial platforms. Implemented time synchronization, pose-aware coordinate registration, cross-view data association, and MoE-style uncertainty-aware gating to dynamically weight observations from different UAVs, sensors, and viewing angles based on confidence, distance, occlusion, and image quality. The framework aggregated distributed

UAV detections into a unified global perception state, reducing single-view ambiguity and improving localization robustness under viewpoint variation, partial observability, and communication constraints.

- Published 1 paper, filed 1 patent and registered software copyright.

Lightweight ToF-Based Hidden Camera Detection Oct 2022 – May 2023

Designed a lightweight hidden-camera detection model based on **Time-of-Flight (ToF) imaging**, leveraging active infrared illumination to capture distinctive reflective patterns from concealed pinhole-camera lenses and image sensors. Built a complete **ToF-to-detection pipeline** covering ToF image acquisition, YOLO-format dataset construction, model fine-tuning, candidate bounding-box analysis, confidence-threshold filtering, and area-coverage scanning for mobile and embedded-device deployment. The system identifies concealed cameras by analyzing reflective depth and optical patterns, achieving **0.976 precision**, **0.969 recall**, and **0.976 mAP@0.5**, enhancing privacy protection and optical sensing reliability in real-world edge environments.

- Filed 1 patent and registered software copyright.

High-Precision Pallet Pose Estimation for Autonomous Forklift Operation Jan 2022 – Sep 2022

Developed an RGB-D pallet pose estimation system for autonomous forklift alignment, fusing RGB images and depth sensing to improve spatial perception in warehouse environments; designed an **analytical triangular geometry model** that computes pallet orientation from key-point distance measurements between the camera and pallet edges, enabling accurate pallet insertion and retrieval. Deployed an **8MB** lightweight model on **NVIDIA Jetson Orin NX** with **ROS2 Humble**, achieving **55.8ms** inference latency, approximately **1% distance error at 10m**, and pallet angle estimation within $\pm 1°$, while publishing structured pallet center, distance, and yaw outputs through ROS2 topics.

- Filed 1 patent and registered software copyright.

HopeFOAM Parallel CFD Computing Platform Jan 2018 – Sep 2018

Co-developed HopeFOAM, an open-source parallel CFD computing framework designed as a high-performance alternative to OpenFOAM for large-scale scientific simulation. Conducted research on high-order numerical methods, discontinuous Galerkin solvers, and shock-capturing schemes for compressible and multiphase flow simulations, contributing to scalable numerical modeling and high-performance engineering computation.
https://github.com/HopeFOAM/HopeFOAM

- Published 2 papers and registered software copyright.

Education

2018 – 2022	Doctor of Software Engineering	National University of Defense Technology
2015 – 2017	Master of Mathematics	National University of Defense Technology
2011 – 2015	Bachelor of Applied Mathematics	National University of Defense Technology

Publications

- Enhancing Cross-Modal Person Re-Identification: Multi-Scale Feature Alignment and Optimization [J]
  *Tianyu Zang, **Yunrui Guo***, Pan Wang et. al.*
  The Visual Computer, 2026

- Conditional Variational Capsule Network for Open Set Recognition [C]
  ***Yunrui Guo**, Guglielmo Camporese, Wenjing Yang, Alessandro Sperduti and Lamberto Ballan**
  IEEE/CVF International Conference on Computer Vision (ICCV), 2021

- Learning Capsules for SAR Target Recognition [J]
  ***Yunrui Guo** , Zongxu Pan, Meiming Wang, Ji Wang and Wenjing Yang**
  IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 2020

- A splitting method for the Degasperis-Procesi equation using an optimized WENO scheme and the Fourier pseudospectral method [J]
  ***Yunrui Guo**, Wenjing Yang*, Hong Zhang, Ji Wang and Songhe Song*
  Advances in Applied Mathematics and Mechanics, 2019

- A high order operator splitting method for the Degasperis-Procesi equation [J]
  ***Yunrui Guo**, Hong Zhang*, Wenjing Yang, Ji Wang and Songhe Song*
  Numerical Mathematics: Theory, Methods and Applications, 2018
- A Maximum Principle-Preserving Third Order Finite Volume SWENO Scheme on Unstructured Triangular Meshes [J]
  ***Yunrui Guo**, Lingyan Tang, Hong Zhang and Songhe Song**
  Advances in Applied Mathematics and Mechanics, 2017
- A class of third order finite volume scheme satisfying the local maximum principle on unstructured meshes [J]
  *Lingyan Tang, **Yunrui Guo** and Songhe Song**
  Mathematica Numerica Sinica, 2017
- Stability evaluation of high-order splitting method for incompressible flow based on discontinuous velocity and continuous pressure [J]
  Liyang Xu and Xinhai Xu and Xiaoguang Ren and **Yunrui Guo** and Yongquan Feng and Xuejun Yang*
  Advances in Mechanical Engineering, 2019
- A high order discontinuous Galerkin method based RANS turbulence framework for OpenFOAM [C]
  Liyang Xu, Yuhua Tang, Xinhai Xu, Yongquan Feng and **Yunrui Guo***
  International Conference on Communication and Information Systems, 2017
- A MatLab toolbox for modeling viscoelastic constitutive equation and parameter optimization [C]
  Liyang Xu, Xinhai Xu, Yuhua Tang, Xiaoguang Ren and **Yunrui Guo***
  IEEE International Conference on Computer and Communications, 2017

Skills

AI/ML - Python, PyTorch, TensorFlow, Distributed training, Slurm, TensorRT, vLLM, OpenCV.
Basic - C/C++, Git, LaTeX, Fortran, Matlab, Mathematica.
Other - ROS2, NVIDIA Isaac Sim, MoveIt, CoppeliaSim, NVIDIA Jetson, AI2-THOR, EmbodiedBench, UE4, Airsim.

Languages

Chinese/Mandarin Mother tongue
English Professional proficiency